NEW STRUCTURAL AND ARITHMETIC CONSTRAINTS ON FINITE-SUPPORT PERIODIC HIGHWAYS OF LANGTON'S ANT

ATHARVA JILLHEWAR

Abstract. For the classical two-colour Langton's ant on $\mathbb{Z}^2$ with finite initial black support, we establish structural and arithmetic constraints on the possible *periodic highways*. Our principal results are the following. *(1) An even-winding theorem*: no finite-support pattern can drift to an exact translate of itself with restored heading and zero net change in the number of black cells; consequently every finite-support periodic highway must build a wake whose net black growth is a strictly positive multiple of four. *(2) A signed mod-four wake-residue identity*: the checkerboard-signed strand bases of any periodic highway sum to $2 \pmod 4$ in each residue class of every nonzero drift coordinate, whence the lower bound $g \geq 2\max(|a|, |b|)$ tying growth g to drift (a, b). *(3) A Tait-graph conjugacy*: the ant is exactly conjugate to edge-toggling on a medial (Tait) graph, with a cycle-rank surgery identity $E + \Delta C = 2\beta$. *(4) A two-endpoint collision-chain parity theorem* (and, building on Traldi's interlacement theory, a touch-graph rank computation for the periodic trace). *(5) A two-engine, rank-audited exhaustive computation* excluding every finite-support periodic highway of nonzero drift and period at most 48, so that any nonstandard periodic highway must have even period at least 50. We also give a decidable periodic-realisability criterion with an explicit finite seed, strengthening the forbidden-word program for the ant's trace subshift. The algebraic core of the even-winding and residue theorems is machine-checked in Lean 4. Every result is a necessary condition on a finite-support periodic highway or an exact finite computation; none resolves the (open) convergence conjecture, and we delimit precisely what a solution must still supply.

Date: July 15, 2026.

2020 *Mathematics Subject Classification.* Primary 37B15; Secondary 37B10, 05C10, 68Q80, 68R15.

Key words and phrases. Langton's ant, turmite, cellular automata, highway conjecture, symbolic dynamics, reversible dynamics, medial graph, Tait graph, interlacement.

Contents

1. Introduction

Langton's ant is the turmite on the square lattice $\mathbb{Z}^2$ defined as follows. Each cell is coloured white or black; a pointer, the *ant*, occupies a cell and faces one of the four cardinal directions. At each step the ant reads the colour of its current cell: on white it turns 90° right, on black it turns 90° left; it then flips the colour of the current cell and moves one unit forward in its new heading. Starting from the all-white plane the ant produces, after a transient of 9977 steps, a self-similar diagonal structure—the *highway*—that repeats with least period 104 and translates the ant by $(\pm 2, \pm 2)$ each period.

Throughout we impose the *finite-support* condition: only finitely many cells are initially black. The empirical universality of the highway is the subject of a well-known open problem.

Conjecture 1.1 (Highway conjecture)**.** For every finite initial black set and every initial pose, the orbit of Langton's ant eventually enters a translate, rotation, reflection, and phase shift of the standard period-104 highway.

1.1. Prior art.

The rigorous facts about the classical ant are few. Every trajectory is *unbounded* (Bunimovich–Troubetzkoy [1]; the underlying checkerboard/parity structure is already in Gale–Propp–Sutherland–Troubetzkoy [6]). Prediction problems for the ant are computationally universal and undecidable, so no finite time cutoff can settle Conjecture 1.1 [4]. Gajardo [5] gives an algebraic characterisation of the forbidden factors of the ant's trace subshift. Recent progress concerns *generalised* ants: some rules admit infinitely many highways [3], others have genuinely non-highway emergent behaviour [7], and quantitative confinement bounds are available [2]. We are not aware of prior published results that constrain what a *classical* finite-support periodic highway can look like—a net-growth congruence, drift-versus-growth arithmetic, a translator-exclusion theorem, or a medial-graph analysis for the classical ant. The present paper aims at that structural layer.

1.2. Results.

A *repeating trace* is an infinite turn word w^ω that is realised as the trace of some finite-support colouring (Section 3); its *drift* $d = (a, b)$ is the per-period displacement and its *growth* g the per-period change in the number of black cells. Our contributions, with an explicit account of their novelty:

(1) **Even-winding theorem** (Theorem 6.1): no finite-support clean translator with nonzero drift exists. Hence every finite-support periodic highway has growth $g \in 4\mathbb{Z}_{>0}$ (Corollary 6.5). Bunimovich–Troubetzkoy excludes only *bounded* repeatable orbits; a zero-growth glider is unbounded and is not covered by their theorem, so the strict positivity is new. (The divisibility by 4 alone is the elementary parity Lemma 2.4.)

(2) **Signed mod-four wake-residue identity** (Theorem 7.1) and the lower bound $g \geq 2\max(|a|, |b|)$ (Corollary 7.3). We are not aware of any exact per-residue arithmetic identity, or any linear drift–growth bound, in the literature.

(3) **Tait-graph conjugacy** (Theorem 4.1) and the cycle-rank surgery identity $E + \Delta C = 2\beta$ (Theorem 4.2). We are not aware of prior Langton-ant work using medial/Tait graphs or loop merge/split surgery; the nearest precedent is the weaker checkerboard partition of [6].

(4) **Collision-chain parity** (Theorem 8.1), a two-endpoint $\mathbb{F}_2$ boundary identity for colour defects. We additionally record, building on Traldi's circuit-partition/interlacement theory [9, 8], a rank computation for the periodic "touch graph" (Observation 8.2); a self-contained treatment of the latter is deferred to the companion report [10]. (We say "touch graph" to avoid a clash with the unrelated "interlaced word" of [7].)

(5) **Small-period exclusion** (Theorem 9.1): a two-engine, rank-audited exhaustive search excludes every finite-support periodic highway of nonzero drift and period at most 48. Existing computational work enumerates highways that do occur; we are not aware of a prior exhaustive nonexistence result below a fixed period bound.

Section 3 also gives a decidable periodic-realisability criterion with an explicit finite seed, strengthening the forbidden-word program of [5] from local factor-admissibility to global periodic finite-support realisability; only that strengthening is claimed as new. The elementary heading/black-count invariant (Lemma 2.4) is folklore, implicit in [1, 6].

1.3. **Scope.** Every theorem below is a *necessary condition* on a finite-support periodic highway, or an exact finite computation. None asserts that any orbit converges, and therefore none resolves Conjecture 1.1. What we prove is that any finite-support periodic highway must build a positive wake divisible by four and must satisfy the residue identities and the density bound $g \geq 2\|d\|_\infty$, and that no finite-support periodic highway has period at most 48 (equivalently, any such highway has period at least 50). The single geometric ingredient separating these results from a full solution is stated precisely in Section 10.

2. Setup and elementary facts

A *state* is a triple $s = (B, p, q)$ with $B \subset \mathbb{Z}^2$ finite (the black set), $p \in \mathbb{Z}^2$ (the ant position) and $q \in \mathbb{Z}/4\mathbb{Z}$ the heading, numbered $0 = N$, $1 = E$, $2 = S$, $3 = W$, with unit vectors

$$u_0 = (0, 1), \quad u_1 = (1, 0), \quad u_2 = (0, -1), \quad u_3 = (-1, 0).$$

The *turn* at s is R (right) with $\varepsilon = +1$ if $p \notin B$, and L (left) with $\varepsilon = -1$ if $p \in B$. One step of the dynamics is

$$s = (B, p, q) \;\longmapsto\; (B \triangle \{p\},\; p + u_{q+\varepsilon},\; q + \varepsilon), \tag{1}$$

where $\triangle$ denotes symmetric difference and indices are read modulo 4. The *trace* of an orbit is the infinite word of turns encountered. Since (1) is invertible (step backward one cell, recolour, undo the turn), the dynamics is reversible and orbits never merge. Write $b_t = |B_t|$ and let $\chi(x, y) = (-1)^{x+y}$ be the checkerboard sign.

Proposition 2.1 (Checkerboard/HV partition [6])**.** *The quantity* $(q_t - (p_{t,x} + p_{t,y})) \bmod 2$ *is constant along every orbit. Consequently each cell is entered by the ant always horizontally or always vertically. Calling a cell* vertical *(resp.* horizontal*) accordingly, at a vertical cell the two vertical incident edges are the only admissible arrival edges and the two horizontal ones the only departure edges, and vice versa; every admissible directed lattice edge thus carries a fixed orientation.*

Proof. Each step changes the heading by an odd quarter-turn, so $q_t \bmod 2$ alternates; the displacement $u_{q_{t+1}}$ has odd ℓ^1-length, so $(p_{t,x} + p_{t,y}) \bmod 2$ alternates as well. Their difference is therefore invariant. Fixing a cell z: at every visit the checkerboard parity of z is the same, so $q_t \bmod 2$ is the same, i.e. the arrival axis is fixed. The edge orientations follow because an arrival edge of one endpoint is a departure edge of the other. □

Proposition 2.2 (Alternating visits)**.** *At any fixed cell the encountered turns alternate strictly between* R *and* L*. All but finitely many cells begin with* R*.*

Proof. Each visit flips the cell's colour and the turn is a function of the colour, so the turn alternates. A cell white at its first visit begins with R; only the finitely many initially black cells can begin with L. □

Theorem 2.3 (Unboundedness [1])**.** *Every finite-support orbit visits infinitely many cells.*

Proof sketch. Were the visited set finite, reversibility would make the orbit periodic in the full state. Choose a visited cell z maximising $x + y$; its north and east neighbours are unvisited. If z is horizontal it is entered from the west, and its two successive (white then black) visits leave south and then north, forcing a visit to an unvisited corner neighbour—a contradiction; the vertical case is the 90° rotation. □

Lemma 2.4 (Heading/black-count invariant; folklore). *For every t,*

$$q_t - b_t \equiv q_0 - b_0 \pmod 4, \qquad p_{t+1} - p_t = u_{\,q_0 - b_0 + b_{t+1}}. \tag{2}$$

A heading-resetting period changes b by a multiple of 4.

Proof. A white/R step increments both q and b by 1; a black/L step decrements both. Hence $q_t - b_t$ is constant. Since the move follows the turn, $q_{t+1} \equiv q_0 - b_0 + b_{t+1}$, giving the displacement formula. If $q_N = q_0$ then $b_N - b_0 \equiv 0 \pmod 4$. □

3. A constructive periodic-realisability criterion

We first record the exact test that turns "is w^ω a highway?" into a finite computation. Fix an initial heading q_0 and a proposed turn word $w = w_0 \cdots w_{N-1} \in \{\mathrm{R}, \mathrm{L}\}^N$. Compute the induced path $p_0, \ldots, p_N$ from w (Proposition 2.1 makes each step well defined). Suppose $q_N = q_0$ and $d = p_N - p_0 \neq 0$. Define an equivalence on phases by

$$i \sim j \iff p_i - p_j \in \mathbb{Z}d,$$

and in each class I write $p_i = b_I + a_i d$ for a fixed representative b_I and integers a_i. For $n \geq \min_I a_i$, let $S_{I,n}$ be the subword of all w_i ($i \in I$) with $a_i \leq n$, listed in increasing order of the key $(n - a_i,\, i)$.

Theorem 3.1 (Periodic-realisability criterion). *The infinite word w^ω is the trace of Langton's ant from some finite-support colouring if and only if, for every class I:*

(i) *every $S_{I,n}$ with $\min_I a_i \leq n \leq \max_I a_i$ alternates strictly between* R *and* L*; and*
(ii) *the stabilised word $S_{I,\max_I a_i}$ begins with* R.

When these hold, an explicit finite black seed producing w^ω is

$$B = \{\, b_I + nd \ : \ \min_I a_i \leq n < \max_I a_i \text{ and } S_{I,n} \text{ begins with } \mathrm{L} \,\}, \tag{3}$$

the union over all classes I; the bound $n < \max_I a_i$ together with $n \geq \min_I a_i$ makes B finite and $S_{I,n}$ well defined.

Proof. The occurrence of phase $i \in I$ in period $k \geq 0$ sits at $p_i + kd = b_I + (a_i + k)d$, which equals $b_I + nd$ exactly when $k = n - a_i \geq 0$; chronological order among these occurrences is the lexicographic order of $(n - a_i, i)$. Thus $S_{I,n}$ is precisely the chronological turn sequence at the physical cell $b_I + nd$ under w^ω.

Necessity. Each visit flips the cell, so its turn sequence alternates, proving (i). For $n \geq \max_I a_i$ all phases of I are present and the ordering no longer depends on n: the sequence has stabilised, and the infinitely many cells $b_I + nd$ ($n \geq \max_I a_i$) share this first turn. By Proposition 2.2 only finitely many cells begin with L, so the stable first turn is R, proving (ii).

Sufficiency. Colour exactly the set (3) black. At each boundary cell $b_I + nd$ with $n < \max_I a_i$ the chosen colour reproduces the first symbol of $S_{I,n}$, and strict alternation supplies every subsequent symbol because the cell flips at each visit. At every stabilised cell ($n \geq \max_I a_i$) condition (ii) makes the first encountered turn R, matching its white initial colour. Induction on chronological time shows the ant follows w^ω forever; since $q_N = q_0$ and the position advances by $d \neq 0$ each period, this is a highway. □

Remark 3.2. There is an equivalent single-sequence test: within a class, sort all entries by decreasing a_i, breaking ties by increasing i; every $S_{I,n}$ is a suffix of the result obtained by deleting whole high-level blocks. It therefore suffices to check that this one stabilised word alternates and begins with R, at the cost of one sort per class. To index the classes of $\mathbb{Z}^2/\mathbb{Z}d$ exactly, write $g = \gcd(a, b)$ and $d = g\,(a', b')$ with $\gcd(a', b') = 1$, and choose integers r, s with $ra' + sb' = 1$ (Bézout). Then

$$(\, b'x - a'y,\ (rx + sy) \bmod g \,)$$

is a *complete* invariant of the coset of (x, y): the matrix $\left(\begin{smallmatrix} b' & -a' \\ r & s \end{smallmatrix}\right)$ has determinant $b's + a'r = 1$, so it is unimodular, and in its coordinates translation by $d = g(a', b')$ becomes translation by

$(0,g)$, leaving the first coordinate and the second coordinate modulo g fixed. (The naive residue $a'x+b'y \bmod g$ is *not* complete: for $d=(2,2)$ the cells $(0,0)$ and $(1,1)$ share both $b'x-a'y=0$ and $a'x+b'y \equiv 0 \pmod 2$ yet lie in different cosets, since $(1,1) \notin \mathbb{Z}(2,2)$; the Bézout coordinate $rx+sy$ separates them.) This is the checker used in all computations of Section 9; it strengthens the forbidden-word characterisation of the ant's trace subshift [5] to a decidable test for global periodic finite-support realisability with a constructive seed. A worked instance (the standard highway) is given in Appendix A.

4. The Tait-graph conjugacy

We recast the dynamics on a medial (Tait) graph, exposing topology invisible in the raw colouring. Normalise the HV partition of Proposition 2.1 so that even cells receive vertical arrivals. In the *frozen* routing (the permutation of directed states obtained by ignoring colour flips) the admissible directed arcs form clockwise directed four-cycles around one checkerboard class of unit plaquettes. Label a plaquette by its lower-left corner and take these plaquettes as the vertices of a rotated square lattice. Each ant cell $z=(x,y)$ lies at the crossing of exactly two white four-cycles and corresponds to the *Tait edge*

$$e_z = \begin{cases} \{(x,y-1),(x-1,y)\}, & x+y \text{ even}, \\ \{(x-1,y-1),(x,y)\}, & x+y \text{ odd}. \end{cases} \tag{4}$$

Let $H_B = \{e_z : z \in B\}$ be the finite edge subgraph induced by the black cells.

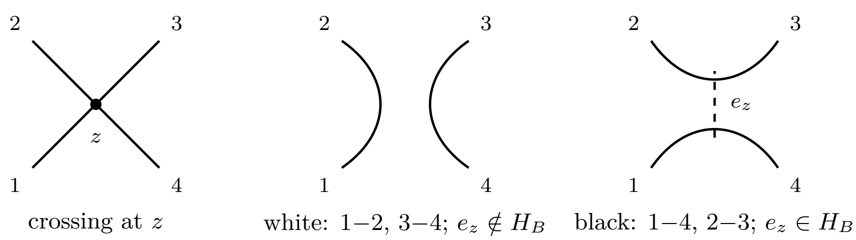

FIGURE 1. The two smoothings of the frozen crossing at an ant cell z. The four incident frozen arcs are labelled $1,2,3,4$. Colour selects which of the two ways they reconnect: white joins 1−2 and 3−4, black joins 1−4 and 2−3. The black smoothing is exactly the occupied Tait edge e_z (dashed).

Theorem 4.1 (Boundary-loop conjugacy)**.** *White and black at z are the two smoothings of the directed crossing at the Tait edge e_z* (Figure 1)*. Consequently:*

(a) *the frozen routing loops are exactly the boundary components of a regular neighbourhood of H_B, together with the untouched four-cycles;*
(b) *the ant advances along its current frozen boundary loop and then toggles the edge e_z; and*
(c) *each step either merges two frozen loops into one or splits one into two.*

Proof. Frozen (colour-ignoring) routing is the permutation ρ of directed edges of the ant lattice sending each directed edge to the next one the ant would follow under a fixed local turn rule; since a white cell turns right and a black cell turns left, at each ant cell z the local routing is one of the two ways of connecting the four directed edge-ends incident to z that avoids a U-turn. These are precisely the two *smoothings* of the 4-valent vertex z of the medial graph, and the Tait edge e_z of (4) records which diagonal pair of plaquette-corners the crossing joins; white is one smoothing and black the other, which is (a) at the level of a single crossing.

Now freeze a colouring B. Away from H_B every plaquette crossing is white, and the white smoothings assemble into the boundary circuits of a regular neighbourhood $\mathcal{N}$ of the edge subgraph

H_B: each occupied edge e_z is thickened to a band, each Tait vertex to a disk, and the frozen routing runs along $\partial\mathcal{N}$, with the untouched white plaquettes contributing the trivial four-cycles. (This is the standard medial/Tait correspondence between a plane graph and the boundary of a ribbon neighbourhood.) This gives (a) globally: the frozen loops are the components of $\partial\mathcal{N}$ plus the untouched four-cycles. Statement (b) is immediate: the live ant sits on the current boundary loop and, on stepping off z, toggles the colour of z, i.e. replaces the smoothing at that single crossing—adding or deleting the band e_z in $\mathcal{N}$.

For (c), toggling e_z is the addition or deletion of one band at a 4-valent vertex, which is a single Dehn-twist-free band surgery on $\partial\mathcal{N}$. Locally it reconnects the two boundary arcs passing the vertex: if the two arcs belong to the same boundary component the surgery splits it into two, and if they belong to different components it merges them into one. There are no other cases, giving (c). □

Let $E = |B|$ be the number of occupied Tait edges, V the number of incident Tait vertices, k the number of nonempty components, and $\beta = E - V + k$ the cycle rank.

Theorem 4.2 (Cycle-rank surgery identity)**.** *A regular neighbourhood of H_B has $k+\beta = E-V+2k$ boundary components. Writing ΔC for the change in loop count relative to the V trivial plaquette loops,*

$$\Delta C = E - 2V + 2k, \qquad E + \Delta C = 2\beta. \tag{5}$$

Adding a bridge merges two loops and fixes β; adding a cycle edge splits a loop and raises β by one; removing a bridge splits a loop and fixes β; removing a non-bridge merges two loops and lowers β by one.

Proof. A regular neighbourhood of a graph with V vertices, E edges, and k components is a compact surface with k pieces and Euler characteristic $V - E$; its boundary consists of $k + \beta$ circles, where $\beta = E - V + k$ is the first Betti number. Formula (5) is the difference from the V trivial plaquette loops. The four surgery cases are the standard effect on the boundary of adding or deleting a bridge (Betti number unchanged; boundary count changes by ∓ 1) versus a cycle edge (Betti number changes by ± 1; opposite merge/split), read off from Theorem 4.1(c). □

Proposition 4.3 (Advanced records are bridges)**.** *Once a bounding-box record lies outside the finite initial support, its first visit is white, and a new east/north/west/south record turns south/east-/north/west. In the Tait graph the corresponding edge has a previously untouched endpoint, hence is a bridge. Therefore sufficiently advanced record creation never raises β; any unbounded production of cycle rank occurs strictly inside already-explored territory.*

Remark 4.4 (No monotone invariant)**.** The active-loop length, its signed area, E, V, k, and β all fail to be monotone, with explicit counterexamples on the blank orbit (e.g. the active loop drops from length 20 to 4 at step 4; the cycle rank takes every value in $\{1, \ldots, 5\}$ during one mature period). Compact states realise both signs of every independent local change of (E, V, k, β), so no nonconstant affine combination of these four quantities is universally monotone. This rules out the most direct potential-function proofs of Conjecture 1.1.

5. Additive charges and clean translators

Definition 5.1. A *clean finite translator with nonzero drift* is a finite state for which, for some $N > 0$ and $d \in \mathbb{Z}^2 \setminus \{0\}$,

$$B_N = B_0 + d, \qquad p_N = p_0 + d, \qquad q_N = q_0.$$

The word *clean* means the entire black set translates: no stationary residue and no newly emitted wake. This is not the ordinary highway, which grows a wake.

5.1. **Laurent charges and axis alignment.** Work in the Laurent polynomial ring $\mathbb{F}_2[X^{\pm 1}, Y^{\pm 1}]$, an integral domain. Let $V_t = 1$ if q_t is vertical and $H_t = 1 - V_t$, and set

$$A_t = \sum_{(x,y)\in B_t} X^x + V_t\, X^{x_t}, \qquad D_t = \sum_{(x,y)\in B_t} Y^y + H_t\, Y^{y_t}.$$

Proposition 5.2. *A_t and D_t are conserved. Hence a clean translator with $d = (a, b)$ satisfies $A = X^a A$ and $D = Y^b D$; since $A(1) = |B| + V$ and $D(1) = |B| + H$ cannot both vanish in $\mathbb{F}_2$ (because $V + H = 1$), the drift coordinates a and b are not both nonzero. Every clean translator is axis-aligned.*

Proof. Conservation of A_t: on a step from a vertical cell, the toggle at column x_t cancels the disappearing vertical correction X^{x_t}; on a step from a horizontal cell, the post-turn motion is vertical, so $x_{t+1} = x_t$ and the toggle cancels the newly appearing correction. Translation covariance then gives $A = X^a A$. In the integral domain $\mathbb{F}_2[X^{\pm 1}, Y^{\pm 1}]$, $a \neq 0$ forces $A = 0$ and $b \neq 0$ forces $D = 0$; but $A(1) + D(1) = 2|B| + 1 = 1 \neq 0$. □

Proposition 5.3 (Additive-charge cocycle)**.** *Let A be an abelian group and $w : \mathbb{Z}^2 \to A$. If there exist heading corrections $F_q : \mathbb{Z}^2 \to A$ (indexed by $q \in \mathbb{Z}/4\mathbb{Z}$) such that $\sum_{z\in B} w(z) + F_q(p)$ is conserved under*

$$F_{q+1}(z + u_{q+1}) - F_q(z) = -w(z) \quad \text{(R)}, \qquad F_{q-1}(z + u_{q-1}) - F_q(z) = +w(z) \quad \text{(L)}, \tag{6}$$

then w satisfies both the four-corner condition

$$w(x{+}1, y{+}1) + w(x{+}1, y) + w(x, y{+}1) + w(x, y) = 0 \quad \text{in } A \tag{7}$$

and $4w = 0$. Conversely, if $A = \mathbb{Z}/4\mathbb{Z}$ (so $4w = 0$ holds automatically) and w satisfies (7), then such F_q exist. Over $\mathbb{F}_2$ the solutions of (7) are exactly $w(x, y) = \alpha(x) + \beta(y)$.

Proof. Necessity. Eliminating the four F_q from (6) around a unit plaquette gives the four-corner condition (7). For $4w = 0$, combine the four relations at a single cell z: the R-at-S and L-at-N relations both express $F_W(z + u_W)$, forcing $F_S(z) = F_N(z) + 2w(z)$, while the L-at-S and R-at-N relations force $F_S(z) = F_N(z) - 2w(z)$; hence $4w(z) = 0$. Thus $w(x, y) = (-1)^{x+y}$ over $A = \mathbb{Z}$ satisfies (7) but admits no such F_q, since $4w \neq 0$.

Sufficiency for $A = \mathbb{Z}/4\mathbb{Z}$. Composing pairs of relations in (6) expresses F_N by its diagonal increments

$$F_N(x{+}1, y{+}1) - F_N(x, y) = -w(x, y) + w(x{+}1, y), \qquad F_N(x{+}1, y{-}1) - F_N(x, y) = w(x, y{-}1) - w(x{+}1, y{-}1), \tag{8}$$

which define F_N on each of the two $x{+}y$-parity sublattices by integration from a basepoint. Modulo four, the circulation of (8) around the elementary diamond $(x, y) \to (x{+}1, y{+}1) \to (x{+}2, y) \to (x{+}1, y{-}1) \to (x, y)$ equals the negative of the sum of the four-corner expressions (7) based at $(x, y{-}1)$ and $(x{+}1, y{-}1)$ (the two differ by $4\,[w(x{+}1, y) + w(x{+}1, y{-}1)] \equiv 0$); hence it vanishes by (7), so the integration is path-independent and F_N is well defined. Now set $F_S = F_N - 2w$ and define F_E, F_W from F_N by (6). Substituting these definitions back into the four R/L relations verifies each case in turn; the only possible ambiguity is between the two ways of reaching F_S—via the R-at-S/L-at-N pair, which gives $F_N + 2w$, and via the L-at-S/R-at-N pair, which gives $F_N - 2w$—and their discrepancy is exactly $4w$, which vanishes. Hence all four F_q are consistent. Over $\mathbb{F}_2$ the solution space of (7) is the separable weights $\alpha(x) + \beta(y)$. □

A mod-four refinement of Proposition 5.2 (using the quadratic solution $Q(x, y) = x^2 + 2xy$ of (7) over $\mathbb{Z}/4$) strengthens the drift of a clean translator to $(4a, 0)$ or $(0, 4a)$, with period divisible by 8 and $N \geq 8|d|$; we use only axis alignment below and refer to [10] for the refinement.

5.2. **The strand decomposition.** Let $g = \#\mathrm{R} - \#\mathrm{L}$ over one period. Lemma 2.4 gives $g \equiv 0 \pmod 4$, and $g \geq 0$ because g equals the number of *odd* stabilised translation classes of Theorem 3.1 (a class is odd iff its stabilised word has odd length, contributing one net black cell per period). We record the endpoint structure used in Section 7.

Proposition 5.4 (Canonical endpoint decomposition)**.** *The one-period signed toggle pattern $D : \mathbb{Z}^2 \to \mathbb{Z}$ (with $D(z) = \#\{\mathrm{R}\ at\ z\} - \#\{\mathrm{L}\ at\ z\}$ over one period) decomposes as*

$$D = S + (T_d - 1)A, \qquad S = \sum_{i=1}^{g} \delta_{c_i}, \tag{9}$$

where $(T_d A)(z) = A(z - d)$, A is a finitely supported moving widget*, and $c_1, \dots, c_g \in \mathbb{Z}^2$ are the bases of the g growing strands, one per odd stabilised class. For the standard word $g = 12$ and $|B_n| = 11 + 12n$ at period boundaries.*

Proof. D is finitely supported (only finitely many cells are visited in one period). Partition the support into translation classes under $z \mapsto z + d$; there are finitely many nonempty classes. Fix a class with base point c and write $f(n) = D(c + nd) \in \mathbb{Z}$, a finitely supported function $\mathbb{Z} \to \mathbb{Z}$. Put $s = \sum_n f(n)$ and $h = f - s\,\delta_0$, so $\sum_n h(n) = 0$. With the shift $(Ta)(n) = a(n-1)$, set

$$a(n) = \sum_{m>n} h(m).$$

Then $a(n-1) - a(n) = h(n)$, i.e. $(T-1)a = h$, so

$$f = s\,\delta_0 + (T-1)a,$$

and a is finitely supported: $a(n) = 0$ for n beyond the support of f on the right, and $a(n) = \sum_m h(m) = 0$ for n far to the left (this is exactly where $\sum h = 0$ is needed). Summing over all classes gives (9) with A the (finite) sum of the class antiderivatives and $S = \sum_{\text{classes}} s_c\,\delta_c$. It remains to identify $\sum_n f$ for a class. Over one period the cells of a class are visited in level order, and the alternating-visit rule (Proposition 2.2) makes the per-class toggle total equal to $+1$ when the class's stabilised word has odd length (an "odd class") and 0 when it has even length; thus $\sum_n f \in \{0, 1\}$ and S is a sum of g unit masses, one per odd class, where $g = \#\{\text{odd classes}\} = \#\mathrm{R} - \#\mathrm{L}$ (the last equality because both count the net black cells created per period). The standard-word figures follow from $|B_0| = 11$ and $|B_{n+1}| - |B_n| = g = 12$. ☐

Corollary 5.5. *If $g = 0$ then a finite stationary set can be deleted to leave a clean finite translator with drift d.*

Proof. Here $S = 0$, so $D = (T_d - 1)A$. Let $b_n : \mathbb{Z}^2 \to \{0, 1\}$ be the indicator of B_n at the n-th period boundary. Since each cell is toggled a net $D(z) \in \{-1, 0, +1\}$ per period (Proposition 2.2) and the period-k toggle pattern is $T_{kd} D$ by translation covariance,

$$b_n = b_0 + \sum_{k=0}^{n-1} T_{kd} D = b_0 + \sum_{k=0}^{n-1} T_{kd}(T_d - 1)A = b_0 + (T_{nd} - 1)A = R + T_{nd}A, \qquad R := b_0 - A.$$

Both R and A are finitely supported, so for n large the supports of R and $T_{nd}A$ are disjoint; since b_n is $\{0,1\}$-valued, R and A are then $\{0,1\}$-valued and $B_n = R \sqcup (A + nd)$. Because the trace is periodic with nonzero drift, each cell is visited finitely often, so at a sufficiently late period boundary the ant never again visits the finite set R. Deleting R there changes no future turn and leaves a state whose black set is $A + nd$, translating exactly by d each period: a clean finite translator. ☐

Sections 6 and 7 prove, by two independent routes, that no such clean translator exists; hence $g \neq 0$.

6. The even-winding theorem

This section proves that no clean translator exists, via a cylinder homology argument. The one step that was only sketched in earlier accounts—the local parity identity relating the turn count at a cell to its horizontal edge parity—is proved in full as Lemma 6.4.

Theorem 6.1 (Even-winding theorem)**.** *No finite-support clean translator with nonzero drift exists.*

For the remainder of the section fix a hypothetical clean translator. Its drift is axis-aligned by Proposition 5.2. A clean translator has $g = 0$, so its period $N = \#\mathrm{R} + \#\mathrm{L} = 2\#\mathrm{R}$ is even; since each step changes $x + y$ modulo 2, the total displacement satisfies $a + b \equiv N \equiv 0 \pmod 2$, so the nonzero drift coordinate is even. After a rotation and reflection write $d = (L, 0)$ with $L > 0$ even—all that the argument below requires. Writing B_j, p_j for the black set and ant position at phase j of one period, reversibility and translation covariance give, for all $n \in \mathbb{Z}$,

$$B_{nN+j} = B_j + nd, \qquad p_{nN+j} = p_j + nd.$$

Quotient the bi-infinite trace by translation through d to obtain a closed walk on the cylinder $C_L = (\mathbb{Z}/L\mathbb{Z}) \times \mathbb{Z}$; because L is even the HV type of Proposition 2.1 descends to C_L. For an undirected cylinder edge, its *multiplicity* is the number of traversals by the projected one-period walk.

Lemma 6.2 (Visit count)**.** *Every vertex v of C_L is visited $n_v = 2k_v$ times in one quotient period, with exactly k_v turns* R *and k_v turns* L*.*

Proof. Let z be a physical representative of v. Each phase j with $p_j \in v + \mathbb{Z}d$ has a unique level m_j with $p_j = z + m_j d$, and phase j in period $-m_j$ visits z; conversely every visit to z over the bi-infinite orbit arises once this way. Hence the visits to v in one quotient period biject with the visits to z over the translated bi-infinite orbit. At phase $nN + j$ the black pattern is the finite translate $B_j + nd$, so z is white for all sufficiently early and all sufficiently late times. Its turn sequence therefore alternates, begins with R, ends with L, and contains equally many of each. ☐

At a vertex v let $H_{v,-}, H_{v,+}$ be the multiplicities of the west and east horizontal edges and $V_{v,-}, V_{v,+}$ those of the south and north vertical edges. By Proposition 2.1 one pair carries all arrivals and the other all departures, and each visit is one arrival plus one departure, so

$$H_{v,-} + H_{v,+} = V_{v,-} + V_{v,+} = 2k_v. \tag{10}$$

Lemma 6.3 (Edge parities)**.** *The horizontal-edge multiplicity has a constant parity $h_y \in \mathbb{F}_2$ along each row y. Every vertical-edge multiplicity is even; write $V_e = 2G_e$ with $G_e \in \mathbb{Z}_{\geq 0}$.*

Proof. Reducing the horizontal half of (10) modulo 2 gives $H_{v,-} \equiv H_{v,+}$, so consecutive horizontal edges in a row share parity; call it h_y. The vertical half gives $V_{v,-} \equiv V_{v,+}$, so vertical parity is constant up each column. The projected walk has finite vertical support, so a vertical edge far enough below or above it has multiplicity 0; constancy then forces every vertical multiplicity even. ☐

Lemma 6.4 (Local parity)**.** *At every vertex $v = (x, y)$ of C_L, $k_v \equiv h_y \pmod 2$.*

Proof. Suppose v is vertical (the horizontal case is the 90° rotation). By Proposition 2.1 the ant arrives at v along a vertical edge—from the south moving north, or from the north moving south—and departs along a horizontal edge. Record each visit by (arrival direction, turn). Using $u_{q+\varepsilon}$ from (1): an arrival from the south (R) departs east, from the south (L) departs west, from the north (R) departs west, from the north (L) departs east. Let $a_\mathrm{R}, a_\mathrm{L}$ count arrivals from the south with the two turns and $b_\mathrm{R}, b_\mathrm{L}$ those from the north. Because the east edge is only ever traversed *out* of v (its other endpoint is horizontal, hence cannot depart along it), its multiplicity equals the number of east departures:

$$H_{v,+} = a_\mathrm{R} + b_\mathrm{L}.$$

The north edge is only ever traversed *into* v, with multiplicity $V_{v,+} = b_{\mathrm{R}} + b_{\mathrm{L}}$. The total turn count is $k_v = a_{\mathrm{R}} + b_{\mathrm{R}}$. Hence

$$H_{v,+} - k_v = (a_{\mathrm{R}} + b_{\mathrm{L}}) - (a_{\mathrm{R}} + b_{\mathrm{R}}) = b_{\mathrm{L}} - b_{\mathrm{R}} \equiv b_{\mathrm{L}} + b_{\mathrm{R}} = V_{v,+} \pmod 2.$$

By Lemma 6.3, $H_{v,+}$ has parity h_y and $V_{v,+}$ is even; therefore $h_y \equiv k_v \pmod 2$. □

Proof of Theorem 6.1. Halving the vertical half of (10) and using $V_e = 2G_e$ gives the exact identity $k_{x,y} = G_{x,y-1/2} + G_{x,y+1/2}$, where $G_{x,y\pm1/2}$ denote the halved multiplicities of the vertical edges below and above v. Fix a column x and sum Lemma 6.4 over all rows:

$$\sum_y h_y \;\equiv\; \sum_y k_{x,y} \;=\; \sum_y (G_{x,y-1/2} + G_{x,y+1/2}) \;\equiv\; 0 \pmod 2, \tag{11}$$

because each interior half-edge G appears exactly twice and the two exterior terms vanish by finite vertical support.

Now cut C_L along a vertical seam between two adjacent columns. The algebraic winding number w of the projected closed walk is the signed number of seam crossings; there is exactly one horizontal seam edge in each row y, of multiplicity $\equiv h_y \pmod 2$, and signs vanish modulo 2, so

$$w \equiv \sum_y h_y \equiv 0 \pmod 2$$

by (11). On the other hand the lift of the closed walk runs from p to $p + (L, 0)$, so it winds exactly once around C_L: $w = 1$. This contradiction proves the theorem for drift $(L, 0)$; reflection and the 90° rotation cover the remaining axis-aligned drifts. □

Corollary 6.5 (Growth is a positive multiple of four)**.** *Every finite-support eventually-translating periodic trajectory has net black growth*

$$g \in 4\mathbb{Z}_{>0}. \tag{12}$$

In particular every finite-support periodic highway builds a nonempty wake.

Proof. Zero drift confines the ant to the finitely many positions of one period, excluded by Theorem 2.3. Negative growth cannot recur forever since $b_t \geq 0$. A heading-resetting period has $g \equiv 0 \pmod 4$ (Lemma 2.4). If $g = 0$ then by Corollary 5.5 the state reduces, after deleting a finite stationary set never revisited, to a clean finite translator with nonzero drift, contradicting Theorem 6.1. Hence $g > 0$, and $g \in 4\mathbb{Z}_{>0}$. □

The standard highway has $g = 12$, consistent with (12). The telescope (11) is the finite algebraic core machine-checked in Lean (Appendix B).

7. The signed mod-four wake-residue theorem

We now give a second, independent proof that no zero-growth periodic highway exists, which additionally pins the arithmetic relation between growth and drift. Recall the strand bases $c_1, \ldots, c_g$ of Proposition 5.4.

Theorem 7.1 (Signed mod-four wake-residue identity)**.** *Let a finite-support repeating trace have restored heading, nonzero drift $d = (a, b)$, growth g, and decomposition (9). Then:*

(i) *if $a \neq 0$, then for every residue $r \in \mathbb{Z}/|a|\mathbb{Z}$,*

$$\sum_{i:\, c_{i,x} \equiv r \pmod{|a|}} \chi(c_i) \;\equiv\; 2 \pmod 4; \tag{13}$$

(ii) *if $b \neq 0$, the analogous sum over each residue of $c_{i,y}$ modulo $|b|$ is $\equiv 2 \pmod 4$;*

(iii) *if $a = 0$, the signed sum $\sum_{i:\, c_{i,x} = x_0} \chi(c_i)$ over each exact column x_0 is $\equiv 0 \pmod 4$; symmetrically for exact rows when $b = 0$.*

The proof rests on a monodromy computation for the potentials of Proposition 5.3.

Lemma 7.2 (Diagonal increment and vertical periodicity)**.** *Fix $a \neq 0$ and any $\alpha : \mathbb{Z}/|a|\mathbb{Z} \to \mathbb{Z}/4\mathbb{Z}$, and set $w(x,y) = \chi(x,y)\,\alpha(x)$, a weight valued in $\mathbb{Z}/4\mathbb{Z}$. Then w satisfies the four-corner condition (7) in $A = \mathbb{Z}/4\mathbb{Z}$, so potentials F_q as in (6) exist, and*

$$F_N(x{+}1, y{+}1) - F_N(x,y) = -\chi(x,y)\big(\alpha(x) + \alpha(x{+}1)\big), \tag{14}$$

while F_N is 2-periodic in y.

Proof. Since $\chi(x{+}1,y{+}1) = \chi(x,y)$ and $\chi(x{+}1,y) = \chi(x,y{+}1) = -\chi(x,y)$, the four-corner sum (7) telescopes to $\chi(x,y)[\alpha(x{+}1) - \alpha(x{+}1) - \alpha(x) + \alpha(x)] = 0$; so F_q exist by Proposition 5.3. Compose the R step at (x,y) heading N (giving $F_E(x{+}1,y) - F_N(x,y) = -w(x,y)$) with the L step at $(x{+}1,y)$ heading E (giving $F_N(x{+}1,y{+}1) - F_E(x{+}1,y) = +w(x{+}1,y)$). Adding,

$$F_N(x{+}1,y{+}1) - F_N(x,y) = -w(x,y) + w(x{+}1,y) = -\chi(x,y)\alpha(x) - \chi(x,y)\alpha(x{+}1),$$

which is (14). The relation $F_S = F_N - 2w$ (from composing the two vertical steps) gives the same increment along $(1,-1)$; combining the $(1,1)$ and $(1,-1)$ increments over a closed square shows $F_N(x,y{+}2) = F_N(x,y)$. $\square$

Proof of Theorem 7.1. Fix $a \neq 0$, α, and w as in Lemma 7.2. Restored heading gives $g \equiv 0 \pmod 4$, so the period is even and $a+b$ is even; hence translation by d preserves χ, and $w(z+d) = w(z)$. Let $\varepsilon = \operatorname{sgn}(a)$, and set $\sigma = \operatorname{sgn}(b)$ if $b \neq 0$ and $\sigma = 1$ if $b = 0$. Move from z to $z+d$ by $|a|$ diagonal steps of vector (ε,σ), followed by the vertical displacement $(0,\, b - \sigma|a|)$, which is even because $a+b$ is even. The uniform diagonal increment of F_N (read off from (14) for $\varepsilon = +1$ and from its reverse for $\varepsilon = -1$) is

$$F_N(x+\varepsilon, y+\sigma) - F_N(x,y) = -\varepsilon\,\chi(x,y)\big(\alpha(x) + \alpha(x+\varepsilon)\big), \tag{15}$$

and χ is constant along the path because $\varepsilon + \sigma$ is even; the vertical part contributes nothing by the 2-periodicity of F_N in y. As x runs once through $\mathbb{Z}/|a|\mathbb{Z}$ each $\alpha(r)$ appears twice, so the increments telescope to $-2\varepsilon\,\chi(z)\sum_r \alpha(r)$, giving the monodromy

$$F_N(z+d) - F_N(z) \;\equiv\; 2\sum_{r\in\mathbb{Z}/|a|\mathbb{Z}} \alpha(r) \pmod 4 \tag{16}$$

(since $-2\varepsilon\chi(z) \equiv 2 \pmod 4$ for either sign of a). The three other headings satisfy the same relation: $F_S = F_N - 2w$ and F_E, F_W differ from F_N by single applications of (6), all of which are themselves d-periodic because w is; so (16) holds with F_q in place of F_N for every q, and for either sign of a.

Charge conservation of $\sum_{z\in B} w(z) + F_q(p)$ over one period gives

$$\sum_z D(z)\,w(z) + F_q(p+d) - F_q(p) = 0, \tag{17}$$

where D is the signed toggle pattern of Proposition 5.4. Because w is d-periodic and $(T_dA)(z) = A(z-d)$, the widget terms cancel: $\sum_z (T_dA)(z)w(z) = \sum_z A(z)w(z)$. Substituting $D = S + (T_d - 1)A$ and the monodromy (16) into (17),

$$\sum_{i=1}^{g} \chi(c_i)\,\alpha(c_{i,x}) + 2\sum_{r\in\mathbb{Z}/|a|\mathbb{Z}} \alpha(r) \;\equiv\; 0 \pmod 4. \tag{18}$$

Taking $\alpha = \mathbf{1}_{\{r\}}$ turns (18) into $\sum_{i:\, c_{i,x}\equiv r} \chi(c_i) + 2 \equiv 0$, i.e. (13), using $-2 \equiv 2$. Rotating by 90° gives (ii). For $a = 0$, take a finitely supported exact-column $\alpha : \mathbb{Z} \to \mathbb{Z}/4\mathbb{Z}$; then b is even, w is d-periodic, the vertical 2-periodicity makes the monodromy vanish, and exact-column indicators give (iii). $\square$

Corollary 7.3 (Strand-density bound)**.** *Every finite-support repeating trace with drift $d = (a,b)$ has*

$$g \;\geq\; 2\max(|a|, |b|). \tag{19}$$

In particular $g = 0$ is impossible—an independent proof of Theorem 6.1. If a residue fiber contains exactly two strand bases, they share checkerboard parity; for $d = (\pm 2, \pm 2)$ and $g = 4$ the 2×2 residue-count matrix is $\left[\begin{smallmatrix} 2 & 0 \\ 0 & 2 \end{smallmatrix}\right]$ or its transpose.

Remark 7.4. The bound (19) is a genuine lower bound but is not claimed to be attained by a chronological highway: the standard highway has $g = 12$ against $2\max(2, 2) = 4$. Equality holds for the static aggregate edge-flow profiles that satisfy (13), but whether any of those profiles is realised by a single ant tour is exactly the kind of chronology question left open in Section 10.

Proof. A sum of m signs ± 1 congruent to $2 \pmod 4$ is nonzero, with m positive and even. Hence each of the $|a|$ residue classes modulo $|a|$ contains at least two strand bases, giving $g \geq 2|a|$; symmetrically $g \geq 2|b|$. If $g = 0$ some class is empty while $d \neq 0$, contradicting (13). A two-element fiber has χ-sum ± 2, forcing equal signs. □

Unlike the even-winding argument, Theorem 7.1 uses neither axis alignment nor the clean-packet reduction; it applies to every repeating trace and additionally fixes the drift–growth arithmetic. Independent computation supports the theorem in the ranges where it is testable (Appendix C): the underlying toggle-charge identity (17) was confirmed on all 172,092 heading-reset nonzero-drift words through length 18. The strand identity (13) and the bound (19) were confirmed on the standard highway, its powers, and its dihedral images; the direct strand-level test is vacuous through length 18 because no realisable nonzero-drift trace occurs in that range.

8. Collision chains and the periodic touch graph

We encode a finite cell set as edges of the bipartite row–column incidence graph: the cell (x, y) is the edge $C_x R_y$ between a column vertex C_x and a row vertex R_y. To a directed state $s = (p, q)$ assign the token $\eta(s) = C_{p_x}$ if q is vertical and $\eta(s) = R_{p_y}$ if q is horizontal.

Theorem 8.1 (Two-endpoint collision-chain parity)**.** *Over $\mathbb{F}_2$, the boundary (set of odd-degree vertices) of the row–column incidence graph of the symmetric difference $B_s \triangle B_t$ equals $\eta(s) + \eta(t)$. Intermediate tokens telescope across any collision checkpoints; consequently at most one defect component of $B_s \triangle B_t$ is non-Eulerian, and a same-pose return ($\eta(s) = \eta(t)$) has only Eulerian defect components.*

Proof. One step toggles the single edge $C_{p_x} R_{p_y}$, changing the degree parity of exactly C_{p_x} and R_{p_y}; the pre- and post-step tokens η differ by the one incidence-vertex whose axis matches the heading, so the boundary contribution of a single step is $\eta(s) + \eta(s')$. Summing over a run telescopes to $\eta(s) + \eta(t)$. A non-Eulerian component needs a nonempty boundary, so at most one exists, and it is absent when $\eta(s) = \eta(t)$. □

This yields a four-cell lower bound on completed lifetimes. Take s immediately after an R visit to $z = (x, y)$ and t immediately before its next (L) visit. Then z is black at both times, so its edge $C_x R_y$ is absent from $G := B_s \triangle B_t$; and since one state is horizontal at z and the other vertical, the tokens are the two ends of that edge, $\eta(s) + \eta(t) = C_x + R_y$. By Theorem 8.1, $\partial G = C_x + R_y$, so C_x and R_y are joined in G by a path avoiding the edge $C_x R_y$; in the simple bipartite incidence graph such a path has at least three edges. Adjoining $C_x R_y$ closes a cycle of length at least four, so the completed R-to-L lifetime involves at least four distinct cells.

We close the section with an observation that packages the periodic trace as a 4-regular Euler system, so that Traldi's circuit-partition/interlacement theory [9, 8] becomes available. We state it as an observation rather than a theorem because a fully self-contained development of the touch-graph formalism (temporal vertices, H/V smoothing circuits, the wake-edge subgraph, and the rank of the boundary cycle space) is lengthy; that development, with complete definitions and proofs, is given in the companion report [10], and we use here only its numerical output on the standard trace.

Observation 8.2 (Periodic touch-graph rank)**.** Pair every stabilised R with its following L into a 4-valent temporal vertex; the g wake R events become degree-two H/V switches, whose suppression yields a connected 4-regular Euler system. With m the number of paired lifetimes, c the number of closed H/V smoothing circuits, and κ the number of cycles in the 2-regular wake-edge subgraph W, the labelled boundary touch graph T and its wake-contraction T/W satisfy

$$\dim Z(T) = m - c + 1, \qquad \dim Z(T/W) = m - c - \kappa + 1,$$

fitting $0 \to Z(W) \to Z(T) \to Z(T/W) \to 0$, and the lifetime-interval rows span $Z(T/W)$ with codimension at most κ in $Z(T)$. On the standard trace one computes $m = 46$, $c = 5$, $\kappa = 1$, projected rank 41, and full boundary rank 42.

Thus ordinary circuit-nullity closes the contracted graph but does not by itself supply the κ-dimensional wake-cycle restriction: at fixed $g = 4$ an explicit abstract family (given in [10]) has $c \to \infty$ and unbounded contracted rank, so no bound depending only on g follows from the graph theory alone. Controlling that kernel with Langton-specific data is one of the open problems of Section 10.

9. Small-period exclusion

All searches are exact: integer coordinates, exact finite sets, and no probabilistic equality tests. Two independently implemented engines are used—a reference implementation of the criterion of Theorem 3.1, and a compiled depth-first search that additionally prunes with Corollary 6.5 and the residue identity of Theorem 7.1. Both engines positively accept the standard word. The methodology, per-shard records, and cross-audit are described in Appendix C.

Theorem 9.1 (No periodic highway of period at most 48)**.** *No finite-support periodic highway of nonzero drift has period at most* 48*. Consequently any nonstandard finite-support periodic highway has even period at least* 50 *(the standard highway has period* 104*; periods* $50, 52, \ldots, 102$ *are not addressed here).*

Verification. Odd periods restore no heading (an odd number of quarter turns is never 0 mod 4), and Corollary 6.5 excludes zero growth at every period, so it suffices to search even periods for positive-growth traces. The evidence is:

(a) a complete enumeration of the legal trace tree through period 32 (46,185,421 feasible nodes over 32 prefix shards; zero highways);
(b) exact positive-growth searches at periods $34, 36, 38, 40$, visiting respectively 9,726,176, 19,781,050, 41,972,884, and 201,924,597 nodes, with zero highways;
(c) two-engine, rank-by-rank audited searches at periods $42, 44, 46, 48$. At the largest, period 48, both engines visited the identical 8,677,026,370 nodes and 451,962,870 leaves over all 2^{16} prefix ranks with no node cap; the reference engine applied the exact criterion of Theorem 3.1 to every leaf, and the residue-pruned engine applied it only to the survivors of Theorem 7.1. Both returned zero certificates, and their per-rank node, leaf, and common legality-pruning counters agree; the reference engine's exhaustive outcome is the certificate of nonexistence, and the residue engine is a cross-check. (At period 46 the two engines likewise agreed on 3,463,441,745 nodes and 98,568,824 leaves with zero certificates.)

Table 1 summarises. Together these exclude every finite-support periodic highway of nonzero drift and period at most 48. □

Proposition 9.2 (Hamming isolation)**.** *Every period-*104 *word distinct from the standard one and within Hamming distance four of it fails the criterion: all* 5,356 *distance-two and* 4,598,126 *distance-four mutations are excluded. Odd Hamming distance cannot restore the heading.*

By [4], no finite period bound can settle Conjecture 1.1; Theorem 9.1 is therefore a lower bound on the minimal period of any nonstandard highway, not a proof of the conjecture.

Period	Method	Nodes	Highways
≤ 32	complete legal enumeration	46,185,421	0
34	positive-growth	9,726,176	0
36	positive-growth	19,781,050	0
38	positive-growth	41,972,884	0
40	positive-growth	201,924,597	0
42	two-engine, rank-audited	528,451,911	0
44	two-engine, rank-audited	1,319,080,456	0
46	two-engine, rank-audited	3,463,441,745	0
48	two-engine, rank-audited	8,677,026,370	0

TABLE 1. Exhaustive exclusion of finite-support periodic highways of nonzero drift. Each entry is complete over its parameter range with no node cap; the period-46 and 48 rows are cross-validated by two independent engines whose per-rank counters agree.

10. The remaining gap

The results above constrain the arithmetic and topology of any finite-support periodic highway, but do not force convergence. Two honest obstructions delimit the gap.

Problem 10.1 (Bounded retreat / bounded active core)**.** Show that for every finite-support initial configuration there is a time after which the ant stays within a bounded corridor behind its record frontier. A stronger Tait-graph formulation that would also suffice is that the active boundary component at or ahead of the current record has uniformly bounded size and that components shed behind an advancing frontier are never reabsorbed.

Under either formulation the frontier window has finitely many states modulo translation, determinism forces eventual periodicity, and Theorem 3.1 classifies the resulting cycle. We do not claim these are the only possible routes to a solution, but no proof along them is known, and the following show why several easy attempts fail: unboundedness (Theorem 2.3) permits an orbit that keeps enlarging a two-dimensional region; none of the natural scalar quantities or affine combinations of (E, V, k, β) considered in Remark 4.4 is monotone; and a local pumping argument is invalid because equal local cores may connect through arbitrarily distant explored territory, turning a later merge into a split (Theorem 4.1).

Problem 10.2 (Periodic classification)**.** Show that every positive-growth periodic trace admitted by finite support is the standard period-104 trace up to symmetry and phase.

At fixed growth $g = 4$ the residue skeletons of Theorem 7.1 and the touch-graph rank of Observation 8.2 leave unbounded free parameters, and explicit heading-reset countermodels satisfy every incidence and residue condition yet fail the cross-period criterion. A resolution therefore needs a single-tour chronology/interlacement theorem valid for unbounded period—genuinely more than the incidence data used here. A *disproof* of Conjecture 1.1 would instead exhibit an infinite certificate: a nonstandard periodic word passing Theorem 3.1, or a finite self-enlarging Tait gadget whose state recurs with a parameter increased. No such certificate has been found.

11. Concluding remarks

We have developed a structural and arithmetic theory of the classical finite-support periodic highway: a decidable realisability criterion, an exact medial-graph conjugacy with surgery calculus, two independent proofs that no zero-growth clean translator exists (so every highway grows a wake of size $\in 4\mathbb{Z}_{>0}$), the strand-density bound $g \geq 2\|d\|_\infty$, a two-endpoint collision-chain parity theorem (and a touch-graph rank computation), and an exhaustive exclusion of all periodic highways of period at most 48. The algebraic cores of the even-winding and residue theorems are machine-checked in Lean 4. The outstanding difficulty is isolated as a bounded-retreat / bounded-active-core lemma together with a single-tour chronology theorem; these, and not the arithmetic constraints proved here, are what a full solution of Conjecture 1.1 must still supply.

Appendix A. Worked example: the standard highway

The blank orbit first enters its repeating trace at step 9977. Normalised so that the ant faces north at the origin and the trace begins with R at a global minimum of the black-count walk, the period-104 word (with R =R, L =L) is

```
RRRRLLRLLRRRRLLRRRRLLRLRRR
RLRLLLLRRRRLRRLRRRRLLLLRLR
RRRLRRRRLLLLRLRRRRLRLLRRLL
LLRRLLRRRRLLRRLRLLRLLRLRLL
```

(read as one word of 104 symbols, four rows of 26). It has $\#\mathrm{R} - \#\mathrm{L} = 12$ and normalised drift $(2, -2)$, in agreement with Corollary 6.5 ($g = 12 \in 4\mathbb{Z}_{>0}$) and Corollary 7.3 ($12 \geq 2\max(2, 2) = 4$). Applying the seed construction (3) of Theorem 3.1 to this exact word (in this exact phase and orientation) produces the 11-cell black seed

$$(-2,-2),\ (-2,-1),\ (-1,0),\ (0,1),\ (1,-2),\ (1,1),$$
$$(2,-2),\ (2,0),\ (3,-1),\ (3,0),\ (4,0),$$

which, with a north-facing ant at the origin, produces the exact 104-step word above immediately, restoring the north heading and translating by $(2, -2)$; the leading corridor is white and the gateway pattern recurs translated by the drift each period. (Both the word and this seed were verified by direct simulation. A different but equally standard normalisation—the phase obtained by cyclically shifting this word by 100 symbols, whose trace begins with L and has drift $(-2, 2)$—is generated by a 13-cell seed; the two must not be interchanged, as they are different phases of the same highway.)

Appendix B. Lean 4 formalization

A dependency-light Lean 4 project (pinned to Lean 4.32.0, standard library only) machine-checks the exact transition kernel and the algebraic cores of the theorems above. Every audited theorem depends only on the standard axioms `propext`, `Classical.choice`, and `Quot.sound`; there are no `sorry`s, no custom axioms, and no `native_decide`. The checked statements include the finite-support step kernel and its toggle lemmas, the heading/black-count reset of Lemma 2.4, the four-corner admissibility and mod-four fiber arithmetic of Theorem 7.1 (including "one residue identity per class $\Rightarrow g \geq 2n$"), the checker-cycle monodromy $2\sum\alpha$ of Lemma 7.2, translated-widget cancellation, and the collision-chain endpoint parity of Theorem 8.1.

The finite parity telescope (11)—the combinatorial heart of the even-winding theorem—is checked in the following self-contained form (`evenWindingParityCore`); `xorAll` is the $\mathbb{F}_2$ sum of a list, `incidences` takes successive XORs, and the two `false` endpoints are the vanishing exterior half-flows:

```
def bxor : Bool -> Bool -> Bool
  | false, b => b
  | true, false => true
  | true, true => false

def xorAll : List Bool -> Bool
  | [] => false
  | b :: bs => bxor b (xorAll bs)
```

```
def incidences : List Bool -> List Bool
  | [] => []
  | [_] => []
  | a :: b :: rest => bxor a b :: incidences (b :: rest)

-- XOR of all adjacent incidences of a list bracketed by two ‘false‘
-- endpoints is ‘false‘: the seam-crossing count is even.
theorem evenWindingParityCore (interior : List Bool) :
    xorAll (incidences (false :: interior ++ [false])) = false := by
  rw [xorAll_incidences, lastFrom_append_singleton]; rfl
```

The formalization isolates, as an explicit interface, the endpoint normal form and lifted potential geometry of a genuine translator, so Theorem 7.1 is certified conditional on a structured trace certificate rather than end to end; neither it nor Conjecture 1.1 is asserted as an unconditional Lean theorem. The project and a build script accompany the paper [10].

Appendix C. Computational methodology and reproducibility

The periodic search space is the tree of legal prefixes. Up to cyclic phase, every positive-growth repeating trace has a representative beginning with R (a realisable trace restarted at a later state remains realisable, and a positive-growth word contains an R), so it suffices to search words beginning with R; by Corollary 6.5 only positive-growth heading-reset words need be considered. To distribute a period-N search, the half-open interval of length-k prefix ranks $[0, 2^{k-1})$ is partitioned into disjoint worker intervals; each worker performs a depth-first enumeration of its subtree, applying the exact criterion of Theorem 3.1 at every leaf. A complete exclusion requires that every rank be covered exactly once, that every shard report completion with no node cap, and that the aggregate node, leaf, and prune counters equal the sums of the per-rank counters.

Two independent engines are run for the period 42–48 certifications. The *reference engine* applies the exact criterion of Theorem 3.1 to every positive-growth leaf and makes *no* use of Theorem 7.1; its zero-hit outcome is therefore unconditional on the residue theorem. The *residue engine* explores the identical prefix tree but first rejects each leaf violating Theorem 7.1, applying the exact criterion only to the survivors. Because both engines enumerate the same tree and reach the same leaves, their node counts, leaf counts, and common legality-pruning counts are identical per shard by construction; they differ only in the residue-prune and exact-evaluation counts, which the residue engine reports and the reference engine does not use. The exclusion itself rests on the reference engine: it applies Theorem 3.1 to *every* leaf and finds no realisable word, independently of Theorem 7.1. The residue engine is an implementation cross-check: its own zero-hit result, together with the identical node/leaf coverage, confirms that the residue pruning discarded no realisable word. For period 48 both engines reported 8,677,026,370 nodes and 451,962,870 leaves over the 2^{16} prefix ranks; the residue engine evaluated the exact criterion on 12,257,208 survivors after rejecting 439,705,662 leaves.

The residue theorem was audited separately, with the following counts over all $\sum_{L\le 18} 2^L = 524{,}286$ turn words of length at most 18: 174,762 restore the heading; 172,092 of those have nonzero drift; and *none* of those 172,092 passes the realisability criterion of Theorem 3.1, consistent with the period-≤ 48 exclusion. The residue theorem (13)–(19) is a statement about *realisable* traces, so its direct test is vacuous in this length range; what the enumeration does verify on all 172,092 words is the underlying additive toggle-charge identity (17) (which holds for every heading-reset word, realisable or not). The strand-level identity (13) and the bound (19) are instead confirmed on the standard word, its powers, and its dihedral transforms. (In particular a word such as RL, with drift $(1, 1)$ and growth 0, restores the heading and lies among the 172,092 but is *not* realisable and is excluded from the strand-level test, as it must be.)

The engines' source, an SHA-256 manifest of every released artifact, the aggregation and cross-audit scripts, and the exact commands to regenerate every shard are publicly available in the accompanying repository [10] at the tagged release `v1.0`; the searches were run on a multi-core cluster with one worker per shard. *(All records are exactly regenerable from the released sources by the commands given there; readers who wish to reproduce the exclusion need only rerun the reference engine.)*

References

[1] L. A. Bunimovich and S. E. Troubetzkoy. Recurrence properties of Lorentz lattice gas cellular automata. *J. Stat. Phys.* **67** (1992), no. 1–2, 289–302.

[2] K. Etse. How long can the escaping ant be confined? arXiv:2606.26677 (2026).

[3] A. Gajardo, V. H. Lutfalla, and M. Rao. Ants on the highway. arXiv:2409.10124 (2024); *Natural Computing* **24** (2025), 497–509.

[4] A. Gajardo, A. Moreira, and E. Goles. Complexity of Langton's ant. *Discrete Appl. Math.* **117** (2002), no. 1–3, 41–50. arXiv:nlin/0306022.

[5] A. Gajardo. A symbolic projection of Langton's ant. *Discrete Math. Theor. Comput. Sci.* (2003). doi:10.46298/dmtcs.2312.

[6] D. Gale, J. Propp, S. Sutherland, and S. Troubetzkoy. Further travels with my ant. *Math. Intelligencer* **17** (1995), no. 3, 48–56.

[7] V. H. Lutfalla. Sideways on the highways. arXiv:2505.05426 (2025).

[8] L. Traldi. Interlacement in 4-regular graphs: a new approach using nonsymmetric matrices. arXiv:1204.0482 (2012).

[9] L. Traldi. Circuit partitions and signed interlacement in 4-regular graphs. arXiv:1607.04233 (2016).

[10] A. Jillhewar. Langton's ant: structural constraints, exact searches, and a Lean formalization (supplementary code, data, and a companion technical report). Zenodo, 2026. DOI: 10.5281/zenodo.21381637. Archived from the GitHub repository `https://github.com/Atharva12456/langtons-ant-highway`; the DOI resolves to the latest archived release, and `results/artifact_hashes.json` in each release gives a SHA-256 for every file.